2 A ball is projected from point A, as shown on Figure 2.1.

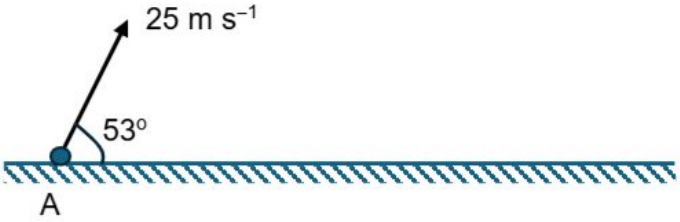


Figure 2.1

The initial velocity of the ball is 25 m s^{-1} at an angle of 53° to the horizontal. The mass of the ball is 0.20 kg.

(a) At a point P on its way to its highest point, the ball's velocity is at an angle of 45° to the horizontal.

(i) Show that the magnitude of the vertical component of the velocity of the ball at point P is 15 m s^{-1}.

[1]

(ii) Calculate the time it takes for the ball to travel from point A to point P.

time = ... s [1]

(b) Calculate the kinetic energy of the ball at a height of 8.0 m.

kinetic energy = ... J [2]

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(c) On Figure 2.2, sketch the variation with height of the kinetic energy of the ball.

No numerical values are needed.

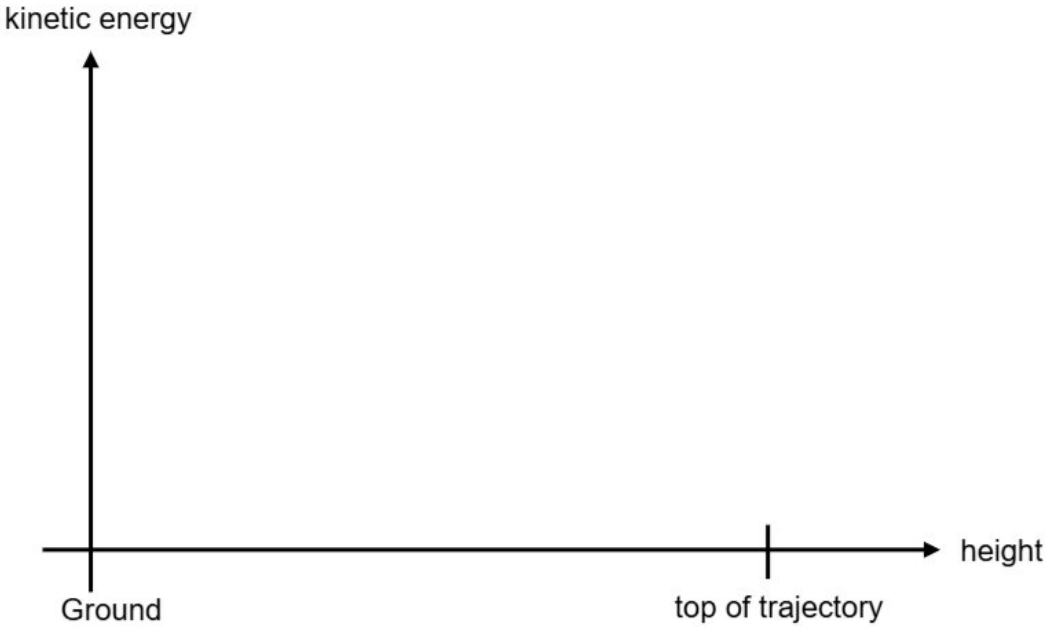


Figure 2.2

[2]

[Total: 6]